

Landsat and Sentinel-2 Based Flood Mapping of the September 2024 Central European Flood Event

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PROJECT SCOPE

Aim of the project

This project aims to develop a machine learning-based approach for assessing floods using satellite imagery, with a focus on the Central European flood event of September 2024.

Leveraging multispectral optical imagery from the Landsat and Sentinel-2 satellite series—including RGB and near-infrared (NIR) bands—the project seeks to estimate flood probability at the pixel level, rather than relying on traditional binary classification methods.

Motivation & Problem Statement

Traditional flood detection methods usually rely on binary classification (flooded vs. non-flooded), providing no indication of prediction confidence.

This project is motivated by the need to go beyond deterministic outputs and develop a model that estimates flood likelihood at the pixel level measured with the help of uncertainty estimation techniques (Monte Carlo dropout).

Research Questions

- Q1: To what extent can high-quality waterbody masks be generated using NDWI and Sentinel-2 Scene Classification Layer (SCL) data?
Q2: How accurately can the proposed deep learning model identify water-covered areas in satellite imagery?
Q3: To what extent can a model trained on Sentinel-2 data be transferred to Landsat imagery?
Q4: Can uncertainty estimates be used to improve the reliability of flood classification?

METHODOLOGY

1. Data Collection & Preparation



- Sentinel-2
- Landsat

Extract images

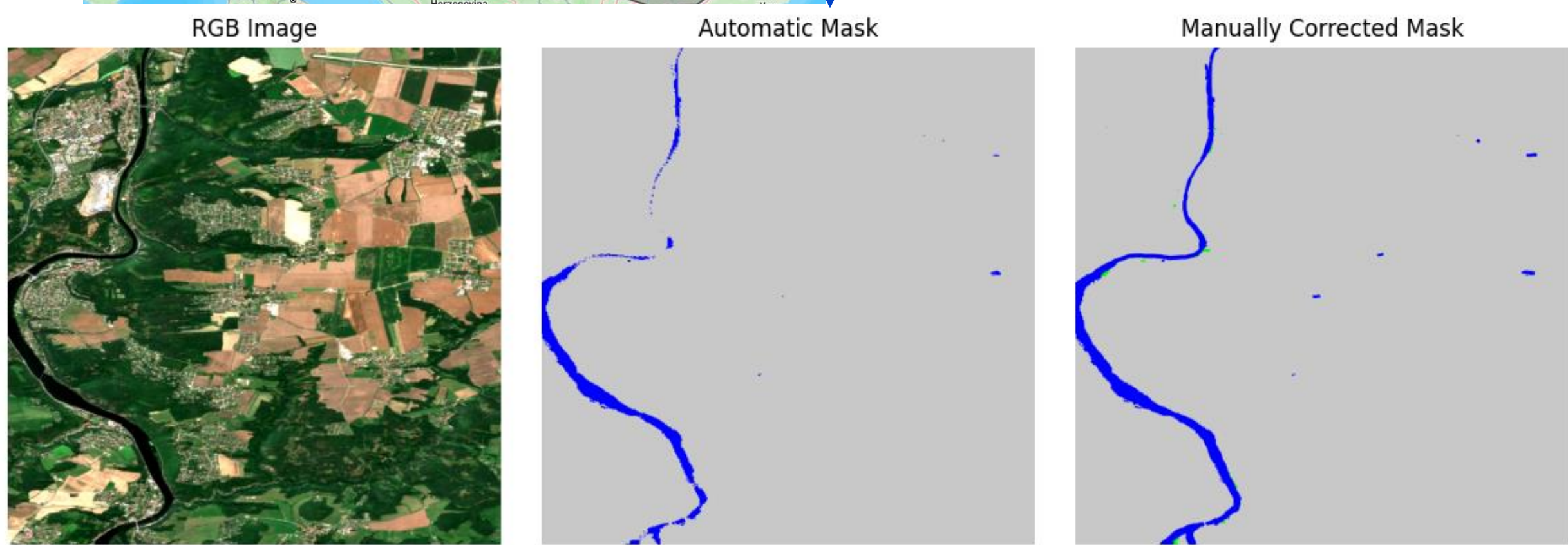
Create & Correct waterbodies masks

Split images into 256x256 tiles

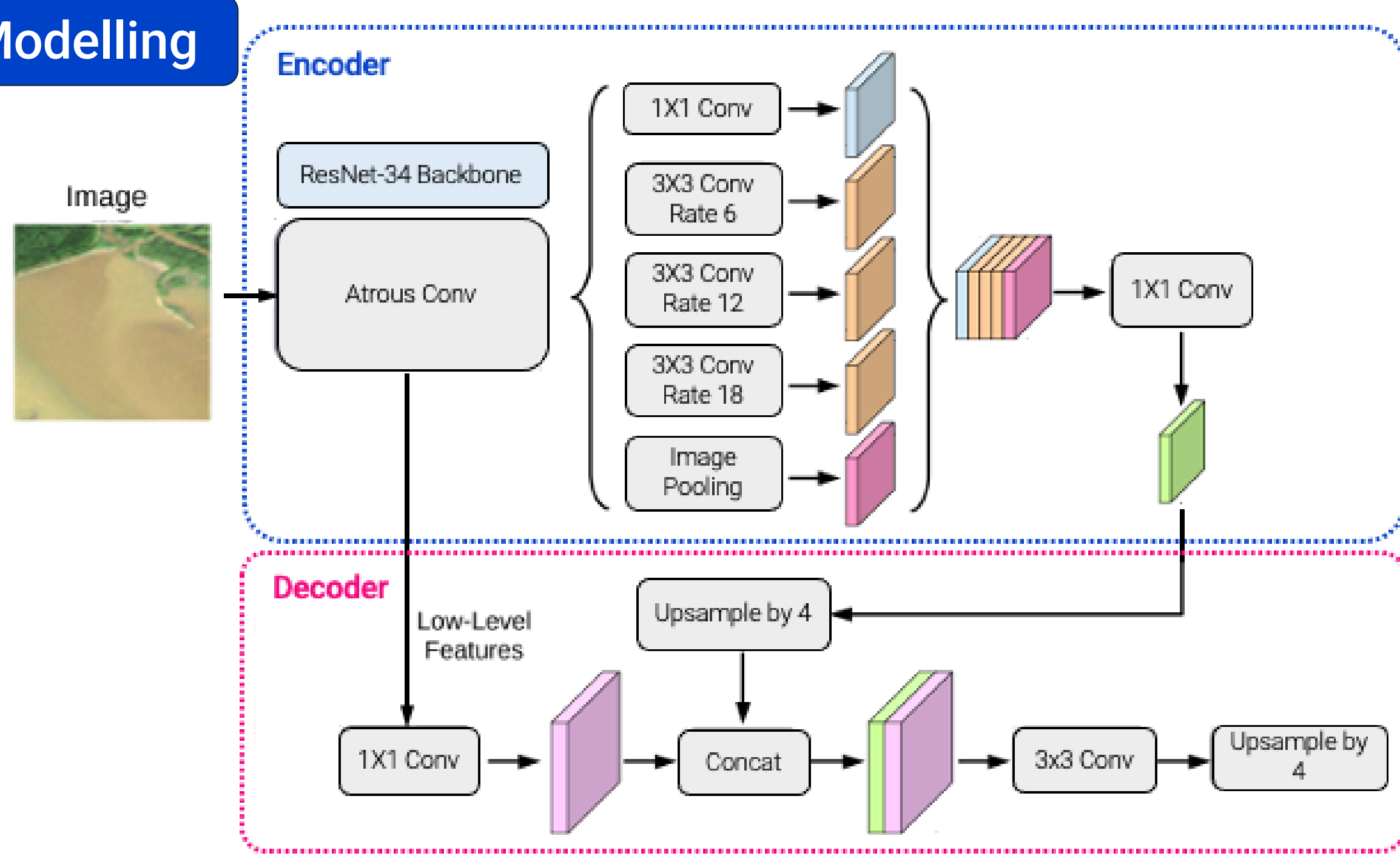
Oversample images with waterbodies

Split train (70%) / test (15%) / validation (15%)

Final dataset: 17,000 samples water - 10% of all pixels



2. Modelling



Final Model Choice
DeepLabv3+ with ResNet34 Encoder

Metrics:

- Precision
- Recall
- F1 Score
- Intersection over Union (IoU)

3. Evaluation

Evaluated on both Sentinel-2 & Landsat sets to test transferability

4. Outcomes

Results for Boris flood event showing uncertainty maps created using Monte Carlo Dropout technique.

* Results are shown for class 1 indicating the quality of water pixels detection

Case 1: Sentinel-2 based test set

Metric	Value
Precision	0.839
Recall	0.898
IoU	0.766
Dice Coefficient	0.867

These results related to research question Q2, showing that the proposed model is effective at identifying water-covered areas in Sentinel-2 imagery.

Case 2: Boris flood based test set

Metric	Value
Precision	0.743
Recall	0.760
IoU	0.602
Dice Coefficient	0.751

These results show that the model is less effective at capturing waterbodies during flood events (specifically, floods caused by storm Boris).

Case 3: Landsat based test set

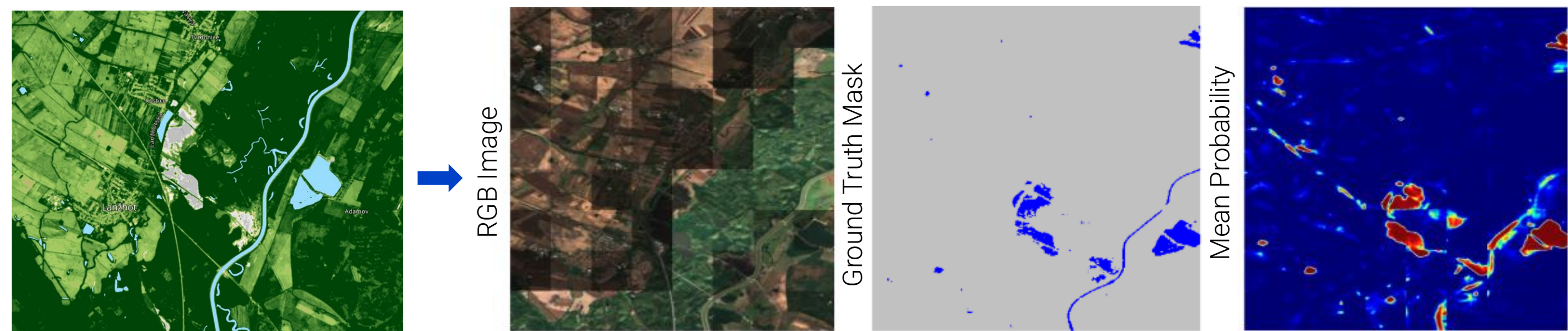
Metric	Value
Precision	0.024
Recall	0.161
IoU	0.021
Dice Coefficient	0.042

This evaluation related to research question Q3, showing that the sharp decline in all evaluation metrics indicates that cross-sensor generalization is highly limited. The model, trained exclusively on Sentinel-2 data, struggles to interpret Landsat inputs.

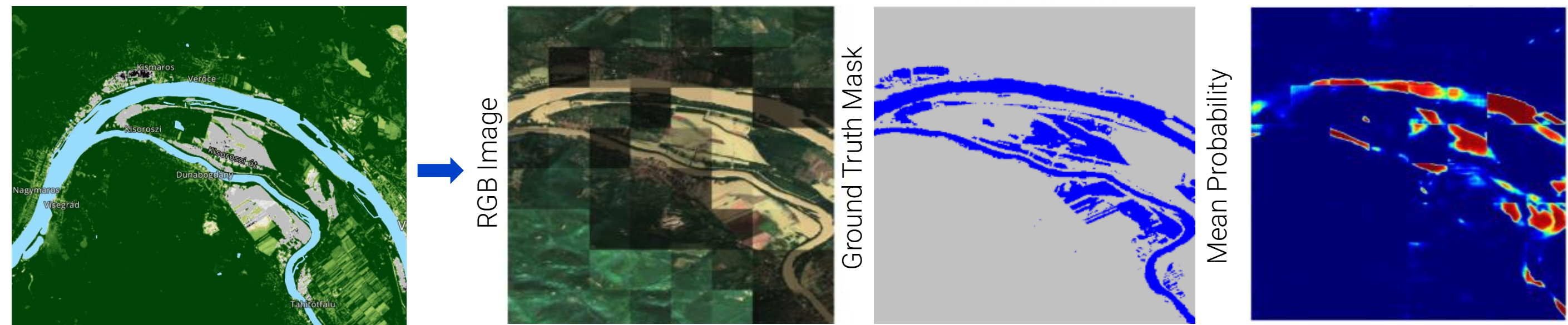
RESULTS

Case 4: Sentinel-2 flood water

Flood water (grey) near Lanžhot (CZ) on 21.09



Flood water (grey) near Kisoroszi (HU) on 20.09



The model detects major flooded areas with high confidence, but struggles to capture smaller areas correctly due to visual changes in floodwaters not well represented in the training data.

Q4: MC Dropout uncertainty helps improve prediction quality. Optimal confidence thresholds (~0.5-0.75) balance precision and recall.

CONCLUSION

Q1: With around three-quarters of the auto-generated masks being of sufficient quality NDWI and SCL-based methods proved to be a viable starting point for semi-automated label generation in waterbody segmentation tasks.

Q2: The model showed good performance on the Sentinel-2 based test set, meaning that the proposed architecture is effective at identifying water-covered areas in Sentinel-2 imagery.

Q3: The model showed poor performance on the Landsat based test set, meaning that trained exclusively on Sentinel-2 data model, struggles to interpret Landsat inputs, which differ in spatial resolution, spectral characteristics, and surface reflectance properties.

Q4: It was shown that by using confidence maps generated during MC Dropout procedure, the quality of generated masks can be improved by achieving a better balance between precision and recall.